

Bayesian calibration of a Cahn-Hilliard surrogate model

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Abstract

Block copolymer (BCP) thin films spontaneously assemble into nanoscale structures with applications in semiconductor and energy materials. [2] Within the Multifaceted Mathematics for Predictive Digital Twins (M2DT) collaboration, models of different fidelities are combined to enable efficient prediction and design of complex physical systems. Connecting these models requires methods for inferring uncertain parameters of lower-fidelity models from high-fidelity simulation or experimental data. This work develops a Bayesian calibration framework for a physics-based surrogate model of BCP self-assembly, using molecular dynamics (MD) simulations as high-fidelity calibration data and a Cahn-Hilliard phase-field model as the computationally efficient surrogate. Fourier-based quantities of interest (QoIs) provide a common representation for comparing morphologies produced by the two models. Gaussian process emulators approximate the mapping from Cahn-Hilliard parameters to these QoIs, enabling efficient Bayesian inference without repeatedly solving the governing partial differential equation. Posterior distributions over the model parameters are sampled using the No-U-Turn Sampler algorithm to identify parameter regions consistent with the MD observations while quantifying calibration uncertainty. The framework provides a basis for uncertainty-aware calibration between models of different fidelities within M2DT. Further development will focus on improving parameter identifiability, the computational efficiency and robustness of the forward-model, posterior-sampling components of the calibration framework, and further investigation of informative QoIs for better parameter identification.

1 Introduction

The Multifaceted Mathematics for Predictive Digital Twins (M2DT) collaboration addresses this challenge by developing hierarchies of interconnected models with different levels of physical fidelity

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and computational cost. Within the BCP application, phase-field models based on the Cahn–Hilliard free energy functional provide a lower-cost, physics-based representation of self-assembly [?]. Rather than resolving individual polymer chains, the Cahn–Hilliard model represents the morphology through an evolving concentration field. However, the effective parameters governing the phase-field model do not have a known direct correspondence to those of the MD system. Before the lower-fidelity model can be used predictively within the model hierarchy, its parameters must therefore be calibrated against high-fidelity observations. This calibration can be formulated as an inverse problem: given observations of a physical system, infer the model parameters capable of producing behavior consistent with those observations. A conventional optimization approach could identify a parameter set that minimizes a chosen discrepancy between the Cahn–Hilliard and MD outputs, but a single optimum does not characterize uncertainty in the inferred parameters. Limited calibration data, imperfect correspondence between models of different fidelities, and weak sensitivity of observable quantities to particular parameters may allow many parameter combinations to remain consistent with the observations. Bayesian calibration instead represents the unknown parameters probabilistically and updates their distributions using the observed data. For the Cahn–Hilliard parameter vector $\boldsymbol{\theta} = (m, \epsilon^2, \gamma)$ and MD-derived observations $\mathbf{y}_{\text{MD}}$, the objective is therefore to characterize the posterior distribution $p(\boldsymbol{\theta} \mid \mathbf{y}_{\text{MD}})$, rather than identify only a single best-fit parameter vector. The shape and concentration of this posterior also provide information about practical parameter identifiability: parameters that are strongly informed by the observations become more constrained, while weakly informed parameters may retain substantial uncertainty.

Direct Bayesian calibration introduces an additional computational challenge because evaluating the posterior requires many evaluations of the forward model. Although the Cahn–Hilliard model is substantially less expensive than MD, repeatedly solving the governing partial differential equation throughout posterior sampling remains costly. In this work, an ensemble of Cahn–Hilliard simulations is therefore used to train Gaussian process emulators that approximate the relationship between the model parameters and Fourier-based quantities of interest (QoIs). The QoIs provide a common low-dimensional representation for comparing structural features of the MD and Cahn–Hilliard morphologies, while the emulators make repeated model evaluations feasible during Bayesian inference.

This work develops the resulting computational workflow for Bayesian calibration of the Cahn–Hilliard surrogate to MD data within the M2DT model hierarchy. The workflow combines efficient numerical simulation of the Cahn–Hilliard model, Fourier-based morphological characterization, Gaussian process emulation, and posterior sampling using the No-U-Turn Sampler. Beyond constructing the calibration framework, the posterior distributions provide a means of examining which Cahn–Hilliard parameters can be identified from the available MD observations and which morphological QoIs provide information about those parameters. This establishes a foundation for continued development of uncertainty-aware model calibration within M2DT, including improved parameter identifiability, forward-model efficiency, and posterior inference.

2 Methods

2.1 Molecular Dynamics Calibration Data

The high-fidelity data used for calibration were obtained from coarse-grained molecular dynamics (MD) simulations of block copolymer thin-film self-assembly performed by the Center for Functional Nanomaterials (CFN) at Brookhaven National Laboratory using LAMMPS. These simulations resolve polymer interactions at substantially higher physical fidelity than the Cahn–Hilliard surrogate model and are therefore treated as the reference data for calibration.

Rather than comparing the MD and Cahn–Hilliard concentration fields directly, structural quantities of interest (QoIs) are extracted from the simulated morphologies. These scalar descriptors provide a common representation for comparing the two models and serve as the observations in the Bayesian calibration problem described in Section ???. The construction of the QoIs is described in Section ??.

2.2 Cahn–Hilliard Surrogate Model

Block copolymer self-assembly is modeled at lower fidelity using a continuum phase-field description based on the Cahn–Hilliard free energy functional with a nonlocal Ohta–Kawasaki interaction term [1]. The phase field $c(\mathbf{x}, t)$ represents the local composition of the two polymer species at position $\mathbf{x} = (x, y, z)$ and time t , with prescribed spatial mean m . The free energy is

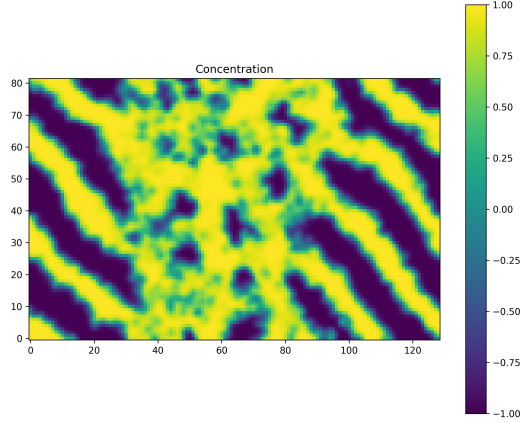


Figure 1: Representative concentration field from a coarse-grained molecular dynamics simulation of block copolymer thin-film self-assembly.

$$E[c] = \int_{\Omega} \left[\frac{1}{4}(c^2 - 1)^2 + \frac{\epsilon^2}{2} |\nabla c|^2 \right] dV + \frac{\sigma}{2} \int_{\Omega} (c - m)(-\nabla^2)^{-1}(c - m) dV, \quad (1)$$

where the double-well potential favors phase separation, the gradient term penalizes sharp interfaces, and the nonlocal term accounts for long-range interactions. In the parameterization used in this work,

$$\sigma = \frac{1}{\epsilon^2 \gamma^2}, \quad (2)$$

where γ is an interaction-length parameter.

The corresponding Cahn–Hilliard dynamics are obtained from the variational derivative of the free energy and describe the evolution of the concentration field toward lower-energy configurations. Unlike MD, which resolves polymer interactions at the molecular scale, the phase-field model represents the evolving morphology through the continuous field $c(\mathbf{x}, t)$. This reduced description provides a computationally less expensive representation of the self-assembly process while retaining the physical mechanisms represented by the free-energy functional.

The uncertain parameter vector considered in the calibration is

$$\boldsymbol{\theta} = (m, \epsilon^2, \gamma). \quad (3)$$

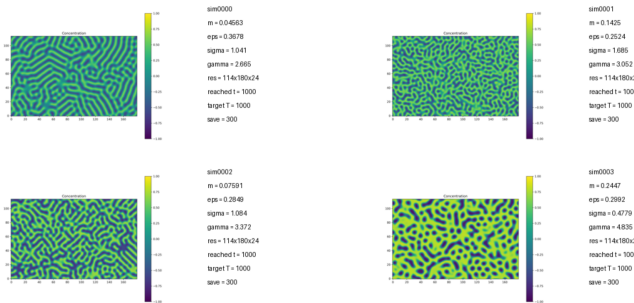


Figure 2: Cahn-Hilliard simulations

The calibration problem is therefore to infer regions of this parameter space for which the Cahn-Hilliard surrogate produces morphological QoIs consistent with those extracted from the MD calibration data.

2.3 Numerical Solution and Simulation Ensemble

We compute the numerical solutions of the governing equation using a first order semi-implicit spectral method implemented in C++. The computational domain is periodic in the x and y directions and satisfies homogeneous Neumann boundary conditions in the z direction. Accordingly, Fast Fourier Transforms (FFTs) are applied in the periodic directions, and a Discrete Cosine transform (DCT) is used in z to satisfy the Neumann boundary conditions.

A semi-implicit time discretization is a better choice than fully explicit methods, because the fourth-order biharmonic operator is numerically stiff. Transforming the governing equation into spectral space diagonalizes the linear differential operators, reducing each timestep to efficient algebraic operations while preserving numerical stability.

2.3.1 Timescales

We used adaptive time stepping that hovered around $dt = 1e - 2$ and $1e - 3$, which was a significant improvement from the backwards difference methods (BDF) with multigrid solvers and fourth order Runge-Kutta solvers that were previously implemented, which required time steps around $dt = 1e - 6$ to $dt = 1e - 9$ respectively, and at which the solve is more computationally expensive at each step. It might take the Runge-Kutta solver around 16 hours to reach a final

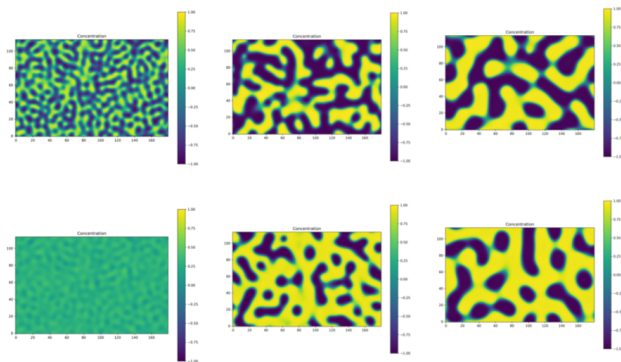


Figure 3: Snapshots of 2 simulations at different times. Column 1: Time = 10. Column 2: Time = 100. Column 3: Time = 1,000

time of 1.5, the BDF multigrid solver 7 hours to reach a final time of 10, and the semi-implicit spectral solver less than 3 hours to reach a final time of 1,000. This was important because the model has different approximated timescales in different areas of parameter space. Parameter sets previously believed to produce negligible evolution had simply not been simulated long enough. When evolved for much longer time, these simulations developed distinct morphologies. It is also important to let the simulation get to an equilibrium state with minimal oscillations. Since we extract Quantities of Interest only from the final state, it is important to integrate long enough for transient oscillations in the system to decay for reliable calibration data.

The characteristic evolution time varies substantially across parameter space, making the ability to efficiently integrate over long physical times particularly important. Spectral methods are also particularly suitable for phase-field models because the concentration field is smooth. It provides rapid convergence while accurately resolving the interfacial structure.

2.4 Morphological Quantities of Interest

Although the molecular dynamics and Cahn-Hilliard simulations are modeling the same system, when comparing large amounts, we need an efficient and effective way to compare two differently parametrized systems, where one is an approximation of the other. Instead, we map each 3D concentration field into a scalar quantities of interest (QoIs) that capture the dominant structural characteristics while providing a common representation for comparing MD and CH simulations. More importantly, we care about the major structures and features in the morphologies, not the

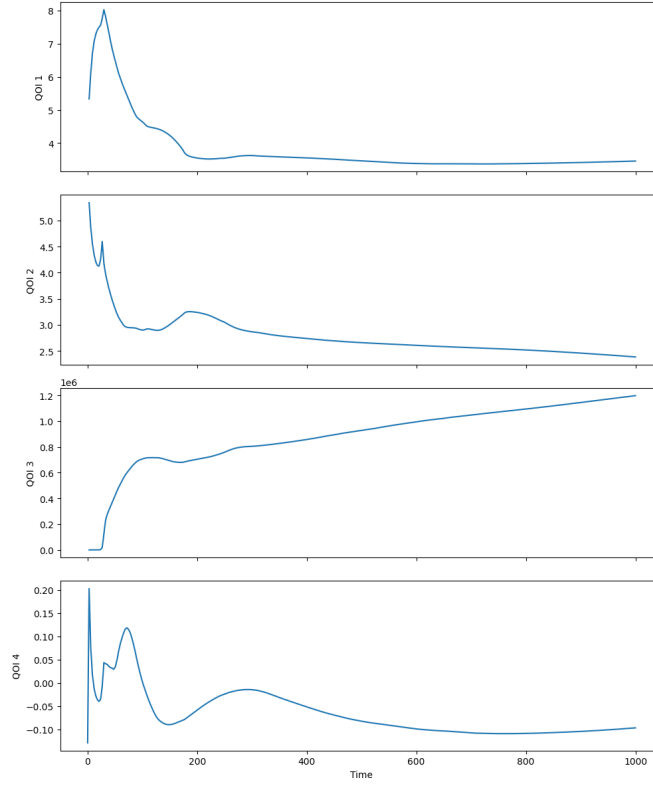


Figure 4: As the concentration field evolves, the QoIs are calculated for each snapshot. Convergence is desirable.

specific pixel concentrations. To capture these main features, we map the domain to a 2D power spectrum using a Fourier transform. This is mapped to a 1D peak, from which we gather scalar values from the location, width, and amplitude of this peak. These 3 QoIs give us information about the characteristic length scale throughout the morphology, spacing between domains, and how organized and repeating the morphology is. We also observe the angular distribution of the Fourier spectrum to get our fourth QoI, which captures information about how well ordered it is in six directions, also called anisotropy. These provide a common space to directly compare Molecular dynamics and Cahn-Hilliard model outputs to calculate residual for the Cahn-Hilliard model.

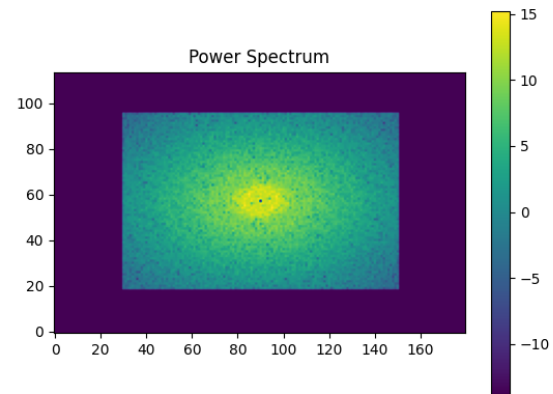


Figure 5: Radially averaged power spectrum, from which QoIs 1,2,3 are derived

2.5 Gaussian Process Emulation

Bayesian calibration requires evaluating the forward model thousands of times to explore the posterior distribution. Although the Cahn-Hilliard model is substantially less expensive than Molecular Dynamics, repeatedly solving the governing partial differential equation during Markov Chain monte Carlo is still computationally prohibitive for thousands or tens of thousands of samples. To make this feasible, we train a Gaussian Process (GP) emulator to learn the mapping of parameters to their QoIs that would be produced by the Cahn-Hilliard and QoI workflow. We train a separate GP emulator on each QoI in the same way. Predictive accuracy was assessed using leave-one-out cross validation of the four QoI emulators to estimate emulator error over the training ensemble. Unlike deterministic interpolation methods, the GP emulator provides both a predictive mean and an estimate of predictive uncertainty. This uncertainty is propagated into the Bayesian likelihood to account for emulator error during calibration. During calibration, the emulator replaces repeated PDE solves while providing an estimate of the uncertainty associated with each prediction. The total observation variance is modeled as

$$\sigma_{total}^2 = \sigma_{GP}^2 + \sigma_{RMS}^2 + \sigma_{reg}^2$$

where σ_{GP}^2 is the GP predictive variance, σ_{RMS}^2 is the leave-one-out RMS emulator error, and σ_{reg}^2 is a small numerical regularization term. Incorporating emulator uncertainty allows Bayesian inference to account for both interpolation error and finite training data.

2.6 Bayesian Calibration

The calibration problem is formulated as a Bayesian inverse problem, where the objective is to infer the Cahn-Hilliard model parameters that produce morphologies consistent with the Molecular Dynamics observations. Rather than identifying a single best-fit parameter set, Bayesian calibration estimates a posterior probability distribution over the parameter space, thereby quantifying uncertainty arising from limited observations, surrogate error, and model discrepancy. Uniform prior distributions were assigned over parameters m, ϵ, γ that produced coherent morphologies in the Cahn-Hilliard model. Assuming independent Gaussian observation errors, the likelihood is constructed from the residual between the predicted and observed QoIs using the total variance

described in Section 2.5. the posterior distribution is obtained through Bayes’ Theorem.

$$p(\theta|y) \propto p(y|\theta)p(\theta)$$

where $p(\theta)$ denotes the prior distribution $p(y|\theta)$ is the likelihood, and $p(\theta|y)$ is the posterior distribution. The posterior therefore identifies regions of parameter space that are most consistent with the observed Molecular Dynamics morphologies while accounting for uncertainty in both the surrogate model and the emulator. Because the posterior distribution cannot be evaluated analytically, samples were generated using the No-U-Turn Sampler, an adaptive Hamiltonian Monte Carlo algorithm that efficiently explores continuous parameter spaces without manual tuning of proposal distributions. Multiple independent Markov chains were run. We ran standard convergence diagnostics, including trace plots, the potential scale reduction and factor $\hat{R}$ effective sample size to diagnose runs.

3 Results

3.1 Gaussian Process Validation

Before performing Bayesian calibration, the Gaussian Process emulator was evaluated using leave-one-out cross validation. Root mean squared error (RMSE) was computed for each quantity of interest to verify that the surrogate accurately reproduced the Cahn-Hilliard simulations. The low prediction errors indicate that the GP emulator provides an accurate approximation of the forward model while reducing the computational cost of evaluating the likelihood during MCMC.

3.2 Markov Chain Monte Carlo with No-U-Turn sampling algorithm

Hamiltonian Monte Carlo using the No-U-Turn Sampler successfully converged for each calibration problem. five independent chains were run with 1,500 warmup iterations, and 5,000 posterior samples per chain. Average acceptance probabilities ranged around 0.80 to 0.90, indicating stable sampling performance by the algorithm. Tree depths remained relatively small, which indicates that the posterior was explored efficiently without requiring excessively long trajectories. The likelihood covariance reports confirmed that the likelihood was modeled using a diagonal covariance matrix, consistent with the assumption that the QoI observation errors are independent.

3.3 Posterior distributions conditioned on individual QoIs

The posterior distributions obtained by conditioning on each QoI separately demonstrate that the four QoIs provide different amounts of information and sensitivity to the model parameters.

3.4 3.3 Molecular Dynamics Targets

Posterior distributions obtained by calibrating to a single MD simulation were compared with those obtained by calibrating to the average of all MD simulations. The posterior shapes were qualitatively similar between the two calibration targets. Major modes remained in the same parameter regions, and the sensitivity of QoIs were largely the same. Averaging the MD simulations reduced random variability in the target observations, but didn't change the inferred parameter distributions, indicating that the Bayesian calibration framework is reasonably robust to the choice of MD target and the dominant source of uncertainty arises from parameter identifiability rather than stochastic variation among MD simulations.

3.5 Parameters

The strongest parameter constraints were observed for ϵ^2 . Conditioning on each QoI showed that smaller values of ϵ^2 were favored, however QoI 4 was the most sensitive and provided the most information. γ largely resembled the uniform prior, indicating that it might be as relevant to a change in morphology, but it is more likely that the QoIs are not sensitive to a change in the γ parameter. This may be because γ seemed to affect the timescale of the system rather than the morphology. Initially, tests were run for average concentration m on the range $[-1,1]$, and a bimodal distribution was consistently sampled, favoring higher absolute values of m . Recall that in the Cahn-Hilliard equation, mass is conserved, so this value is constant throughout the evolution. Then we constrained the sampling range to $[0,1]$, which shouldn't change the distribution since m is average concentration of two phases, so it doesn't matter whether the phase is A or B, because they are treated equally by the simulation. We found that the conditioning on the different QoIs produced different results on the favored values of m , although notably QoI 4 favored values of m close to 0 more heavily, while the other QoIs produced broadly similar distributions. These results suggest that the available QoIs provide only weak information regarding m when considered

individually.

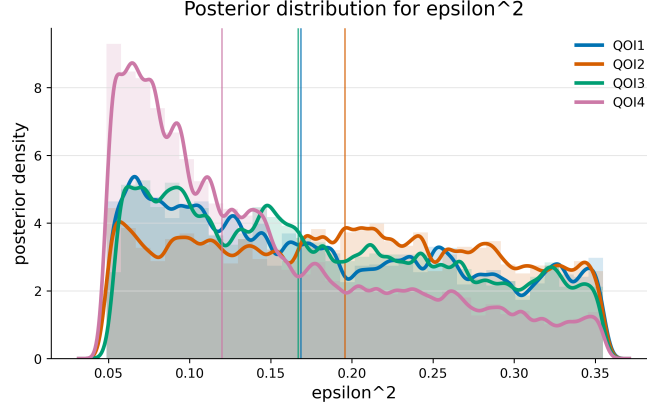


Figure 6: Posterior distribution over ϵ^2

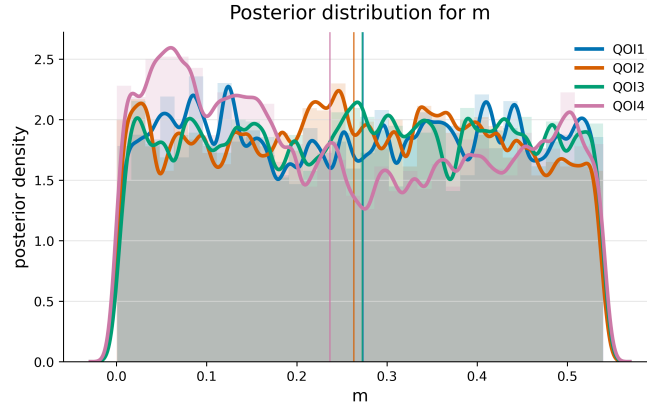


Figure 7: Posterior distribution over m

3.6 Timescale and evolution

We found that the physical rate of evolution depended on several things. The ϵ parameter has a strong influence on the timescale, because it controls the interfacial width, which is the boundaries between distinct domains. In the Cahn-Hilliard partial differential equation, ϵ is on the fourth order biharmonic term/gradient term, so a small ϵ results in larger gradients and thinner interfaces. this often produces faster interfacial relaxation, in other words, the system reaches a stable state faster. The γ parameter is related to $\sigma = \frac{1}{\epsilon^2 \gamma^2}$ which is the coefficient of the nonlocal Ohta-Kawasaki term to account for the long range interactions of the BCP. Since γ is inversely related to σ , a large γ or smaller σ is associated with weaker long-range interactions, which in turn results in slower large-scale ordering. In the same way, smaller γ causes stronger long-range interactions,

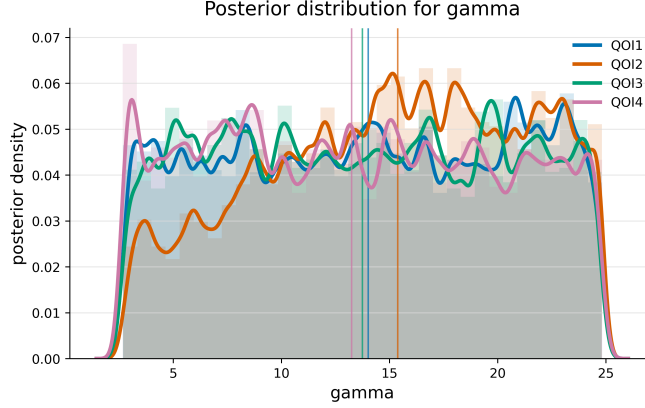


Figure 8: Posterior distribution over γ

which results in faster domain formation. We used random initial conditions to allow the system to undergo spinodal decomposition. If the simulation starts already phase separated, then it may get trapped in an energy well and evolve towards a final state artificially. Adaptive time stepping is advantageous because the system evolves very quickly in the beginning, so a smaller timestep better captures this evolution. As the system's evolution slows down closer and approaches a stable state, the timestep is allowed to increase according to an error threshold.

4 Discussion

The results demonstrate that Bayesian calibration provides considerably more information than a single optimal parameter estimate. The posterior distributions quantify which regions of parameter space remain consistent with the Molecular Dynamics observations and reveal the relative identifiability of each parameter. Although posterior distributions were broadly consistent across different MD targets, some variability remained, which indicate*** One area of further exploration might be finding a better way of computing the residual between the Cahn-Hilliard surrogate model and the calibration data. In other words, finding better Quantities of Interests that are the most sensitive to parameters, therefore they are the most informative when finding posterior distributions. This will help to better distinguish informative regions of parameter space. Although posterior distributions across MD targets largely resembled each other, there is still variation which tells us that the MD simulations aren't perfect either. Another area of further exploration might be exploring how the noise term can be calculated in the likelihood function. There is debate whether

including certain noise terms produce more accurate results.

5 Acknowledgements

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